Indian Institute of Management Calcutta

Programme: MBA

Name of the Course: Cybersecurity for Business

Core / Elective …ELECTIVE (Areas: Management Information systems)

Credit: 1.5

Academic Year & Term: 2026-27, 6th Term

Course Coordinator: Anik Mukherjee

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Instructor: Anik Mukherjee.(Sessions 10 * 1.5 hours)

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Introduction to Course /Course Description

To effectively prepare and respond to the ever-increasing and evolving cyberattacks, organizations must not only have a comprehensive plan but also execute that plan with great precision and consistency. Cybersecurity readiness is a critical and distinctive organizational competency, and this course is designed to enhance managerial awareness and capability to develop and sustain this competency. The course will provide a holistic and comprehensive insight into the different aspects of cybersecurity program management from development to administration, evaluation, and improvement of processes. The course will shed light on various cyber attacks on e-commerce and financial sectors. Students will participate in a simulation to get a feel for cyber-attack related decision-making challenges.

This course will enable the participant to:

- Understand the different types of cyber security vulnerabilities and challenges that plague organizations.
- Recognize the various people-, process-, and technology-driven cyber security defense mechanisms and best practices.
- Understand a defense-in-depth information security strategy.
- Understand the implications of cybersecurity and privacy laws and regulations
- Cybersecurity applications.

Programme Level Learning Goals (7)
Learning Goals **
3. Develop a global perspective that is built upon knowledge of economic, technological, regulatory and political forces that are shaping a new international order in the 21 st century

Learning Objective: **Familiarize with global factors/phenomenon affecting the business scenario**

6. Acquire capacity to apply their professional knowledge and skills to diagnose and resolve business problems in actual organizational settings

a) Acquire the ability to identify organizational problems in both real life settings and through simulated situations
b) Acquire the ability to interpret and apply knowledge of concepts

International Components in your course (if any)
Case studies mentioned in the readings.
Guest lecture.

Pre-requisites
N.A.

Required Text Book(s)

N.A.

Recommended Text Book(s)

- NA

Course Pack Distribution to students

A collection of relevant articles/papers may be provided to the students during the course.

Technology enabled learning component for your course
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Session Plan

Session	Topic	Intended learning outcome	Readings

PART 1: Demystifying Cyber Security			
1.	Cyber Attacks	Types of Cyber Attacks. How to mitigate them. Threat Actors. Types of Threat Actors. Who are they? How do they operate?	• Act, Don't React: A leader's guide to cyber security- *Harvard Business Review Case* (2019) • Cyber Attack - What Are Common Cyberattacks?--by *Cisco*
2.	Cyber Breach Incidents and Lessons Learnt	Cyber security and Data breach	• Cyber Attack at the University of Calgary- *Harvard Business Review Case* (Jan 2021) • Cyber Breach at Target— *Harvard Business Review Case* (Jul 2016)
3.	Security Counter measures	How to detect rough devices? Authentication, Authorization and New Development	• How to prepare your business to survive the next zero-day attack- *Forbes* (August 9, 2022) • CIA Triad
4.	Cyber Security Simulation Game	In-Class Exercise to help students understand how to respond collaboratively in crisis situations and recognize the importance of cyber security readiness.	• IT Management Cyber Attack reading material- *Harvard Business Review Simulation* (2021)
PART 2: Defense in Depth			
5.	Defence in depth	Cyber Security Tech Stack. Layers of security. Zero trust	• Security Stack by CSC • What is VPN? How It Works, Types of VPN by Kaspersky • VPN Meaning: What Is a VPN & What Does It Do? by Avast • What is Zero Trust? by IBM • Evolving Zero Trust by Microsoft
6.	Defence in depth	Inline Security device. Monitoring devices. How VPN works?	• DeepLocker by IBM
PART 3: Applications			
7.	Cyber Security and IoT	Cloud Computing Internet of Things Operational Technologies (OT)	• IoT-Convenience vs Privacy and Secrecy, *Business Horizon* (2015) • Securing IoT using blockchain and zero trust: Block Armour's dilemma—*IIMCCRC Case* (2019)
8.	Guest Lecture	TBA	To be announced later
9.	Cyber Threats in Banking Sectors	Challenges financial sectors face Security measures in banks Threat vectors	• The Cyber Threat Landscape: Confronting Challenges to the Financial System, *Carnegie Working Paper Series*, 2019 • Enduring Cyber Threats and Emerging Challenges to the Financial

			Sector, *Carnegie Working Paper Series*, 2020 • SBI Personal Banking • IBM Safer Payments
10.	Cybersecurity Governance	NIST Framework. Domestic vs International Cyber Laws. Cyber Security Challenges in e-Governance (AADHAR/PAN etc). Cyber Ethics.	• Navigating Cybersecurity Risks in International Trade-*Harvard Business Review* (December 2021) • What the new VPN rules means for internet users in India - Times of India • What India's new VPN rules mean for your privacy - The Economic Times • The Aadhaar Card: Cybersecurity Issues with India's Biometric Experiment - The Henry M. Jackson School of International Studies (washington.edu)

Evaluation Components

✓ Components	✓ Weightage
Class discussion	**20%**
Assignment	**40%**
End Term Exam	**40%**

Online Course Management (moodle)/course web

Pedagogy

The pedagogy will consist of lectures, guest lectures, simulations, case studies and discussions.